# Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

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# Agenda


1. Problem Statement
2. Background
3. System Design & Architecture
4. Methodology
5. Previous Work: Esophageal Varices Detection
6. Implementation
7. Conclusion & Future Work
8. References

# The Healthcare Documentation Challenge

- **35-37%** of physician time spent on paperwork, not patient care [1, 2, 3]
- **1.77 hours daily** on documentation outside office hours [3, 4, 5]
- **75%** of healthcare providers report negative impact on patient care [4, 6, 7]
- **21.2%** absolute reduction in burnout with AI-powered tools [2, 1]

**Solution:** Automate endoscopic report generation using Vision-Language Models

# Vision-Language Models in Healthcare

- Combine computer vision and natural language processing [8, 9, 10]
- Process both visual and textual medical data simultaneously [8, 11]
- Enable automated report generation and diagnostic assistance [8, 12]
- Reduce after-hours documentation time by nearly **1 hour per week** [2, 1]

**Key Models:** CLIP, LLaVA, Flamingo [8, 12, 11, 13]

**Applications:** Radiology, pathology, endoscopy [8]

# Literature Survey: Documentation Burden

| Metric | Citation |
|---|---|
| Physicians spending 1.77 hours daily on docs | [3, 4, 5] |
| EHR Survey | [3, 4, 5] |
| Negative impact on patient care | [4, 6, 7] |
| Time inappropriate/reduces patient time | [4, 6] |
| Documentation only for billing | [4, 7] |

# Endoscopic AI Applications

- **FDA-Approved AI Models for GI Endoscopy:** 13 of 882 (June 2024) [17]
- **Endo-FM Foundation Model** [18]
  - 33,000+ video clips (5 million frames)
  - Superior classification, segmentation, detection
  - Captures local and global dependencies
- **EndoVis Datasets** [19, 20, 21]
  - 2017: 472 QA pairs, 5 videos
  - 2018: 11,783 QA pairs, 14 videos

# Pre-training Strategies for Medical VLMs

- **Two-Stage Curriculum Learning** [14, 15, 12]
  - Stage 1: Medical concept alignment
  - Stage 2: Instruction tuning
- **Domain-Specific Pre-Training** [16]
  - EndoViT: 700,000+ endoscopic images
  - 10% performance gains over ImageNet pre-training
- **Foundational Techniques** [8]
  - Contrastive learning
  - Masked language modeling

# System Architecture Overview

**Three Core Components:**

1. **Vision Encoder** - Processes endoscopic imagery
2. **Language Encoder** - Handles textual medical information
3. **Fusion Module** - Combines multimodal features

**Multimodal Transformer Architecture** [8, 9]

- Specifically designed for endoscopic procedure analysis
- Comprehensive pipeline: videos/images → structured reports
- Leverages visual patterns + contextual medical knowledge

# Figure 1: Custom Dataset Workflow

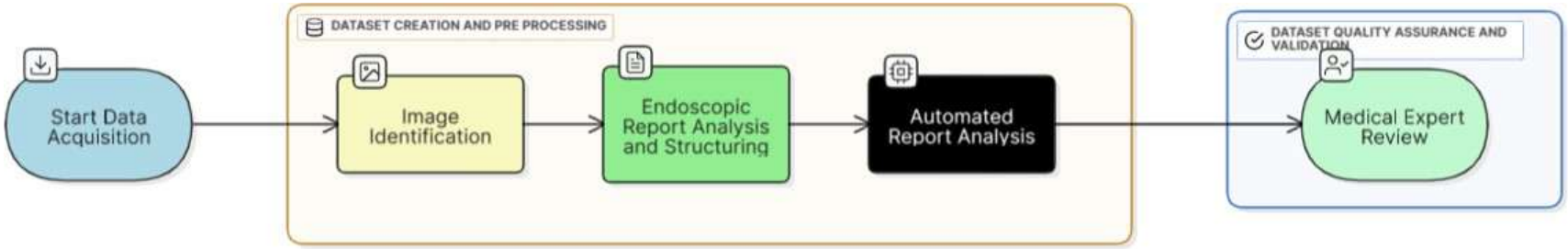



Figure: Data Workflow: From 2,500 patient folders to 32,545 Q&A pairs [22, 23]

# Automated Question-Answer Pair Generation

**Multi-Stage Process:** [24, 25]

1. **Medical Entity Extraction** - Key concepts and diagnoses
2. **Question Template Design** - Diagnostic, anatomical, pathological, procedural, comparative
3. **Automated QA Generation** - Multiple question variations per concept [13, 26]
4. **Answer Synthesis** - Accurate answers from reports + medical knowledge

**Result:** Diverse, clinically relevant question-answer combinations

# Two-Stage Training Framework

**Stage 1: Medical Domain Pre-Training** [15, 14, 12]

- **Objective:** Establish medical concept alignment
- **Base Model:** LLaVA-1.5 with frozen CLIP ViT-
- L/14 **Trainable:** Only multimodal projection layer
- **Datasets:** EndoVis 2017/2018 + Endo700k (700,000+ images) [16, 20, 19]
- **Training:** Contrastive learning between images and descriptions

**Stage 2: Task-Specific Fine-Tuning** [15, 12]

- **Objective:** Instruction tuning for diagnosis and report generation
- **Dataset:** Custom GMC-derived Q&A pairs (32,545 samples)
- **Strategy:** Parameter-Efficient Fine-Tuning (LoRA) [27, 28]
- **Unfrozen:** Projection layer + language model parameters [14, 15]

# Figure 2: System Architecture and Methodology

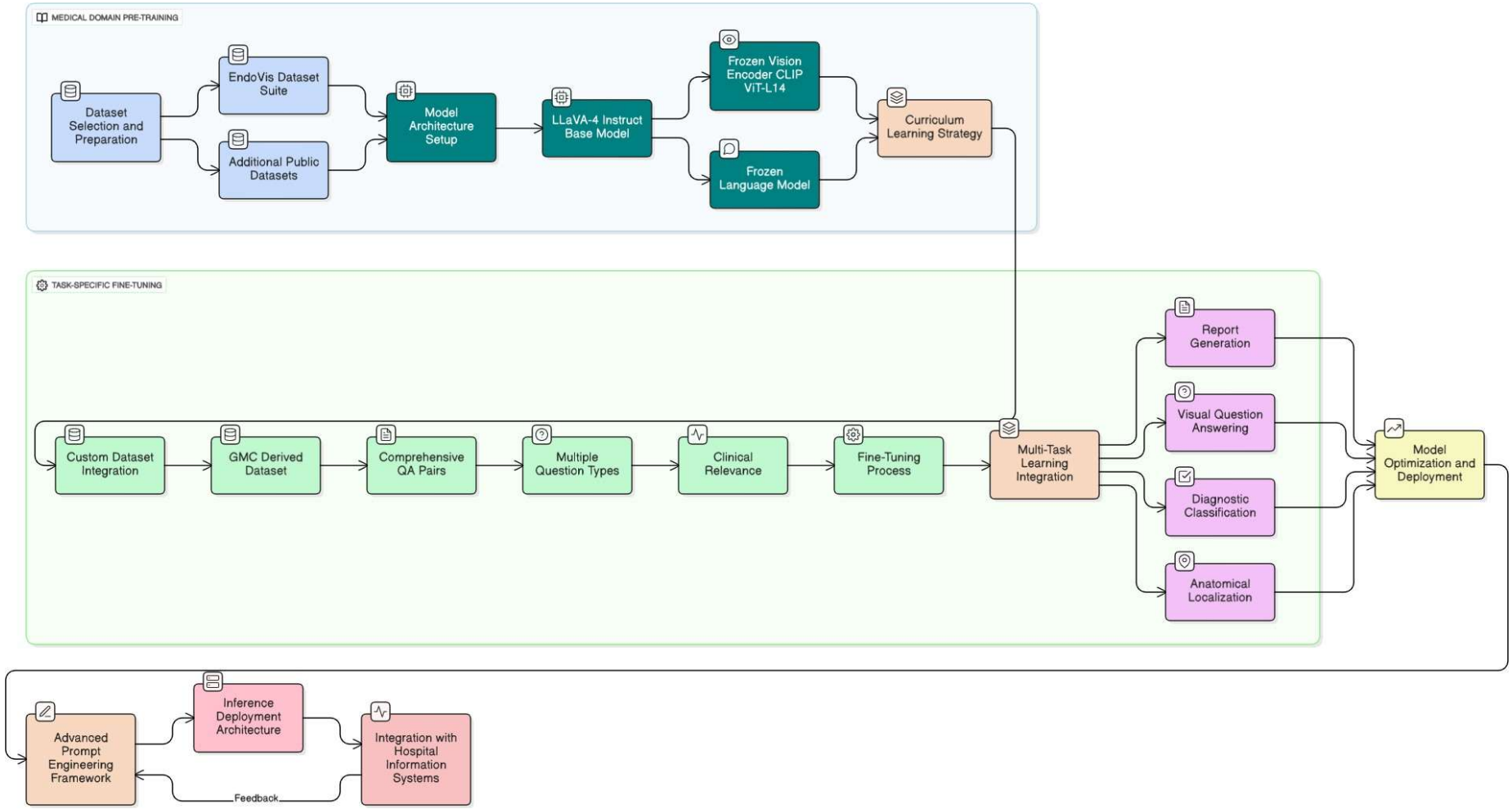



Figure: Complete two-stage training pipeline with data flow and model components [15, 14]

# Model Optimization & Deployment

**GGUF Format Conversion** [29, 30]

- Optimizes CPU and mixed CPU-GPU inference
- Significantly improved model loading times [31, 29]
- Supports CPU-only inference with selective GPU offloading [32, 31]

**Multi-Level Quantization Strategy** [30, 32]

| Format | Use Case | Size Reduction |
|---|---|---|
| Q8_0 | High-end workstations | ~50% |
| Q5_K_M | Standard clinical computers | ~69% |
| Q4_K_M | Basic clinical terminals | ~75% (3.25GB) [33] |
| Q3_K_L | Mobile deployments | Ultra-compact |

# Inference Deployment Architecture

**Two Deployment Strategies:**

1. **vLLM Framework** [34, 35]
   - High-throughput clinical environments
   - Concurrent access scenarios
2. **Ollama Framework** [34, 36, 35]
   - Individual workstations
   - Low-resource environments
   - Resource-constrained deployment

**EHR Integration:**

- RESTful API architecture (secure, authenticated endpoints)
- Real-time synchronization with patient records
- Quality assurance dashboard for performance monitoring

# Automated Data Engineering Pipeline

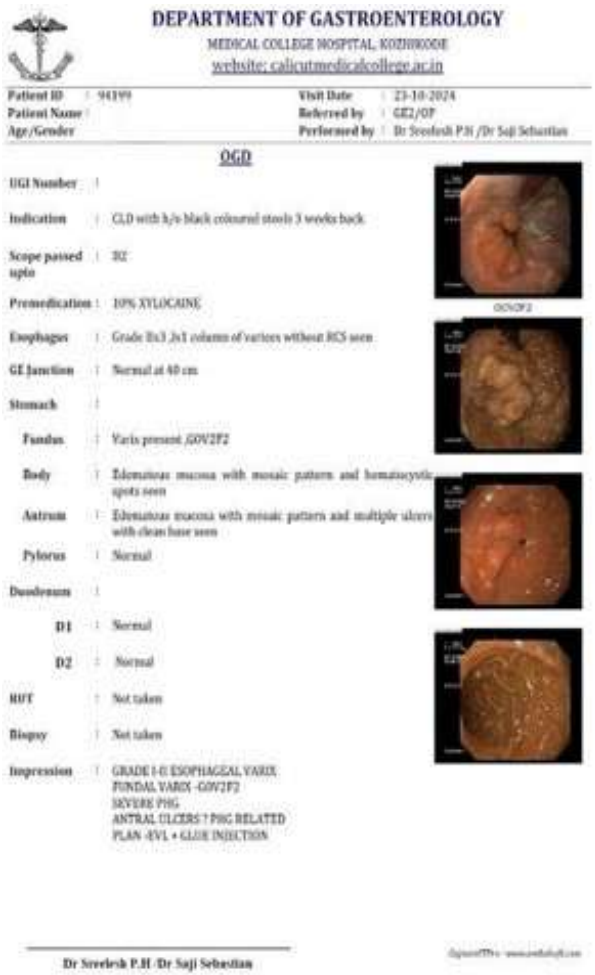
**Challenge:** Unstructured medical PDFs with inconsistent layouts

**Solution:** Custom Python-based pipeline using PyMuPDF

**Two Key Components:**

1. **Positional-Heuristic Extraction**
   - Dynamic boundary detection for variable column widths
   - Multi-block key heuristic for multi-line fields
   - Metadata filtering (exclude non-clinical headers)
2. **Multimodal Instruction Formatting**
   - Conversational transformation of key-value pairs
   - Image consolidation and unique indexing
   - Contextual linking in JSON structure

# Figure 3: Sample Endoscopy Report and Structured JSON Output



DEPARTMENT OF GASTROENTEROLOGY
MEDICAL COLLEGE HOSPITAL, KOZHIKODE
website: calicutmedicalcollege.ac.in

Patient ID : 94199
Patient Name :
Age/Gender
Visit Date : 23-10-2024
Referred by : GE2/OP
Performed by : Dr Sreelesh P.H /Dr Saji Sebastian

OGD

UGI Number :
Indication : CLD with h/o black coloured stools 3 weeks back
Scope passed upto : D2
Premedication : 10% XYLOCAINE
Esophagus : Grade II x3, 1x1 column of varices without RCS seen
GE Junction : Normal at 40 cm
Stomach :
Fundus : Varis present ,GOV2F2
Body : Edematous mucosa with mosaic pattern and hematocystic spots seen
Antrum : Edematous mucosa with mosaic pattern and multiple ulcers with clean base seen
Pylorus : Normal
Duodenum :
D1 : Normal
D2 : Normal
RUT : Not taken
Biopsy : Not taken
Impression : GRADE I-II ESOPHAGEAL VARIX
FUNDAL VARIX -GOV2F2
SEVERE PHG
ANTRAL ULCERS ? PHG RELATED
PLAN -EVL + GLUE INJECTION

Dr Sreelesh P.H /Dr Saji Sebastian

```
{
  "Patient_Clinical_Info": {
    "Indication": "CLD with history of black colored stools 3 weeks back",
    "Premedication": "10% Xylocaine"
  },
  "Procedure": {
    "Scope_passed_upto": "D2",
    "RUT": "Not Taken",
    "Biopsy": "Not taken"
  },
  "Anatomical_Findings": {
    "Esophagus": {
      "Findings": "Grade II x3, 1 cm column of varices without RCS"
    },
    "GE_Junction": {
      "Findings": "Normal"
    },
    "Stomach": {
      "Findings": "Normal"
    },
    "Fundus": {
      "Findings": "Varis present, GOV2F2"
    },
    "Body": {
      "Findings": "Edematous mucosa with mosaic pattern and hematocystic spots"
    },
    "Antrum": {
      "Findings": "Edematous mucosa with mosaic pattern and multiple ulcers with clean base"
    },
    "Pylorus": {
      "Findings": "Normal"
    },
    "Duodenum": {
      "D1": "Normal",
      "D2": "Normal"
    }
  },
  "Diagnostic_Impression": {
    "Esophageal_Varix": "Grade I-II",
    "Fundal_Varix": "GOV2F2",
    "Portal_Hypertensive_Gastropathy": "Severe PHG",
    "Antral_Ulcers": "Likely PHG related"
  },
  "Plan": {
    "Endoscopic_Variceal_Ligation (EVL)",
    "Glue Injection"
  }
}
```


Figure: Real endoscopic report with clinical findings and multimodal instruction format showing images linked to Q&A pairs in JSON structure [22]

# Custom Dataset Creation

**Data Source:** Government Medical College (GMC), Kozhikode [22, 23]

- **Raw Data: 2**,500 patient folders
- **Content:** Semi-structured PDF reports + endoscopic images
- **Final Q&A Pairs:** 32,545 unique samples
- **Data Split:**
  - Training: 30,917 samples
  - Validation: 1,627 samples

**Multimodal Association:** Each sample = endoscopic images + text question + clinical answer

# Report Analysis and Preprocessing

**Medical Entities Extracted:** [22, 23]

- Anatomical structures
- Pathological findings and characteristics
- Diagnostic conclusions and recommendations
- Procedural techniques and instruments
- Patient symptoms and clinical presentations

**Processing Method:**

- Natural language processing techniques
- Extraction of critical medical elements
- Foundation for Q&A pair generation

# Fine-Tuning Configuration Details

**Vision Encoder:** CLIP-ViT-L/14

- Image resolution: 336×336 pixels
- Patch size: 14×14 (24×24 grid)
- Total vision tokens: 577 per image (576 patches + CLS)
- Critical for capturing minute mucosal lesions

**Training Hyperparameters:**

- **Precision:** bfloat16 (BF16)
- **Optimizer:** AdamW with 8-bit quantization
- **Learning Rate:** $2e^{-4}$ with cosine scheduler, warmup 0.1
- **Batch Size:** 2 micro-batch, 4 accumulation steps = 8 global
- **Attention:** Flash Attention 2
- **Epochs:** 2 epochs, 7,730 steps, 253M tokens

# Esophageal Varices Detection System

**System Architecture:**

- Vision Transformer (ViT) based architecture
- Transformer Encoder with self-attention mechanisms
- Patch and Positional Embedding for spatial relationships
- Linear Projection of Flattened Patches

**Preprocessing:**

- Data augmentation (rotation, cropping)
- Addresses variability in endoscopic imaging
- Improves model robustness across clinical scenarios

**Binary Classification:**

- **Varices:** Presence requiring clinical intervention
- **No Varices:** Normal esophageal condition

# Figure 4: Esophageal Varices Detection Architecture

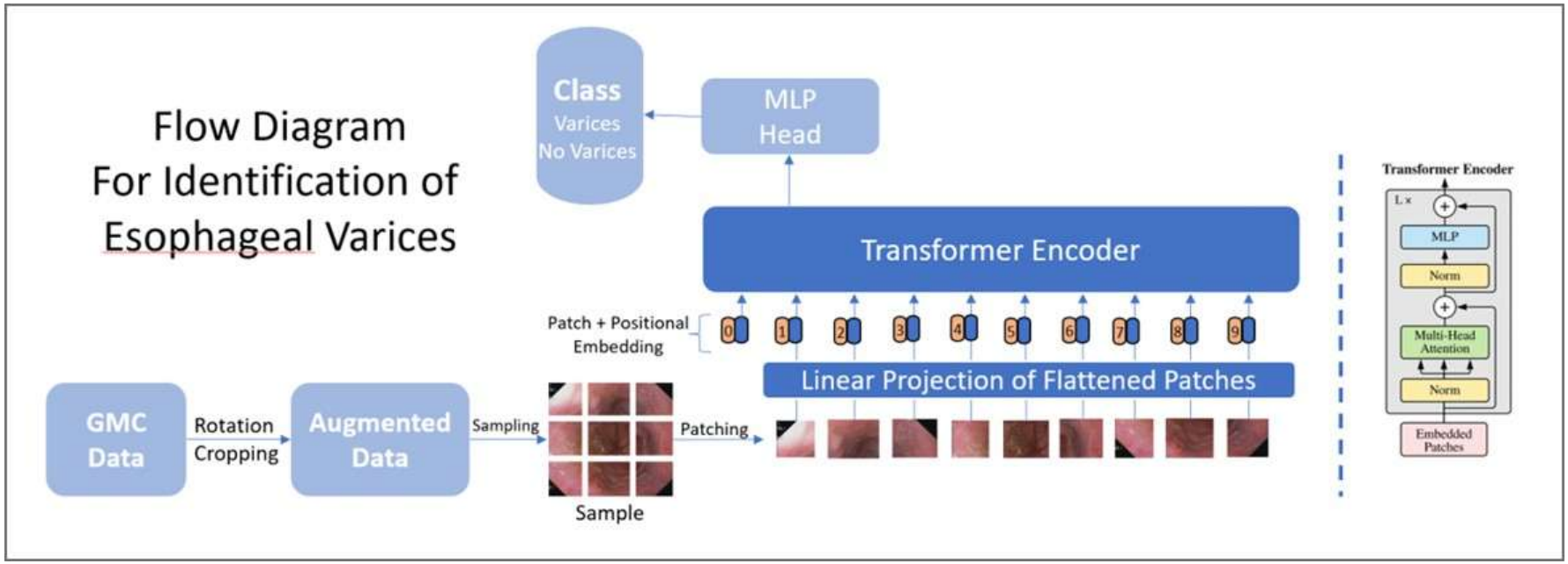



Figure: Vision Transformer-based architecture for automated esophageal varices identification [15, 14]

# Esophageal Varices Detection: Performance

| Metric | ViT-Base | ResNet-18 | ResNet-50 | EfficientNet-B7 |
|---|---|---|---|---|
| Accuracy | 97.14% | 95.00% | 91.43% | 92.85% |
| Precision | 0.9719 | 0.9520 | 0.9151 | 0.9349 |
| Recall | 0.9711 | 0.9500 | 0.9143 | 0.9286 |
| F1-Score | 0.9715 | 0.9497 | 0.9139 | 0.9287 |

**Key Findings:**

- Vision Transformer (ViT-Base): **97.14% accuracy**
- achieved Zero false negatives in evaluation
- ViT outperforms traditional CNNs and hybrid approaches
- Effectiveness of attention-based mechanisms confirmed

# Implementation: Base Model & Technique

**Base Model:** LLaVA-1.5-7B [13, 26]

- Proven capability in general-domain visual instruction following
- Adapted for medical endoscopy applications

**Fine-Tuning Technique:** LoRA (Low-Rank Adaptation) [27, 28]

- Freezes pre-trained model weights
- Injects trainable rank decomposition matrices
- Significantly reduces computational overhead

**Memory Efficiency:**

- bfloat16 precision optimization
- 8-bit quantization with bitsandbytes
- GPU memory footprint reduction

# Conclusion

**Key Achievements:**

- Comprehensive methodology for medical diagnosis and report generation [8, 12]
- Two-stage training approach with domain-specific pre-training [14, 15, 12]
- Advanced quantization (GGUF) for resource-constrained deployment [29, 30]
- Integration capabilities with hospital information systems [22]

**Impact:** [1, 2]

- Significant potential for reducing physician burnout
- Enhanced diagnostic accuracy
- Improved workflow efficiency in clinical settings
- Maintains clinical accuracy on resource-constrained infrastructure

**Path Forward:** Seamless adoption within established clinical workflows

# Acknowledgments


- **Government Medical College (GMC), Kozhikode**
  - Provided endoscopic dataset (2,500 procedures)
  - Clinical expertise and collaboration
- **Medical Staff**
  - Clinical validation and feedback
  - Support during development process


**Thank You!**

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